

Exercise 1: Ranking and Window Functions

	ProductID	ProductName	Category	Price	RowNum	RankNum	DenseRankNum
1	101	Laptop	Electronics	50000.00	1	1	1
2	103	Keyboard	Electronics	1000.00	2	2	2
3	102	Mouse	Electronics	500.00	3	3	3
4	105	Table	Furniture	6000.00	1	1	1
5	104	Chair	Furniture	4000.00	2	2	2

Exercise 2: Aggregation with GROUPING SETS, CUBE, and ROLLUP

	Region	Category	TotalQuantity
1	South	Electronics	8
2	NULL	Electronics	8
3	North	Furniture	3
4	South	Furniture	2
5	NULL	Furniture	5
6	North	NULL	3
7	South	NULL	10

	Region	Category	TotalQuantity
1	North	Furniture	3
2	North	NULL	3
3	South	Electro...	8
4	South	Furniture	2
5	South	NULL	10
6	NULL	NULL	13

	Region	Category	TotalQuantity
1	South	Electronics	8
2	NULL	Electronics	8
3	North	Furniture	3
4	South	Furniture	2
5	NULL	Furniture	5
6	NULL	NULL	13
7	North	NULL	3
8	South	NULL	10

Exercise 3: CTEs and MERGE

	CalendarDate
4	2025-01-04
5	2025-01-05
6	2025-01-06
7	2025-01-07
8	2025-01-08
9	2025-01-09
10	2025-01-10
11	2025-01-11

	ProductID	ProductName	Category	Price
1	101	Laptop	Electronics	50000.00
2	103	Keyboard	Electronics	1000.00
3	102	Mouse	Electronics	500.00
4	105	Table	Furniture	6000.00
5	104	Chair	Furniture	4000.00

Exercise 4: PIVOT and UNPIVOT

	ProductID	1	2	3	4	5	6	7	8	9	10	11	12
1	101	3	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
2	102	5	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
3	104	3	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL
4	105	2	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

	ProductID	MonthNo	Quantity
1	101	1	3
2	102	1	5
3	104	1	3
4	105	1	2

Exercise 5: Using CTE to Simplify a Query

	CustomerID	CustomerName	OrderCount
1	1	Prem	2